# Joao De Campos Carvalho

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## EDUCATION

**University of Toronto** - BASc, Mechanical Engineering + PEY Co-op | Sep 2024 - Apr 2029
Minors: Robotics & Mechatronics, Engineering Business | GPA: 3.25 | CSWA-certified (Dec 2025)

## EXPERIENCE

**Auditorium AI - Operations Lead | Sep 2026 - present, Toronto**

- Built the operating system: shared finance drive, per-event P&L tracker, deals pipeline, gear inventory, weekly founder reports.
- Manage a content lead shipping 8 pieces per month against an approved brief, with weekly check-ins.
- Set up and managed an AI-assisted development workflow for the company web product, coordinating AI-generated code (frontend-heavy) into a reviewed, working codebase.

**Auditorium AI - Lead Design Engineer | May - Aug 2026, Toronto**

- Engineered a spatial-calibration feature mapping automated stage lighting to room geometry, uncovering two silent failure modes every existing check had passed.
- Integrated hardware and software for a single-board-computer field device: field updates, device driver rules, and a 3D-printed enclosure designed in Fusion 360.
- Conducted market and competitive research that redirected the company's target customer segment; produced the follow-up channel strategy.

**UTESCA - VP of Outreach | Sep 2025 - present, Toronto**

- Identify partners at consulting firms with an 80% conversion rate; built an automated outreach tracker in Google Sheets.
- Run weekly prospecting cadence for event partnerships: one consulting-firm contact per week, tracked from first touch to booked meeting.

**UHN Sleep Monitoring Device - Co-Manager | Sep - Dec 2024**

- Co-managed a 6-person team designing a low-cost sleep monitoring device with UHN.
- Designed the CAD model in Fusion 360 and integrated the electronic components.
- Originated 120+ design ideas, converging on a design 30% cheaper than industry competitors.

**Brazilian Robotics Olympics - Team Leader / Chief Engineer | 2021 - 2023**

- Designed and manufactured the robot's custom PCB, circuitry, and rescue vision module.
- Designed and programmed the C# simulation code, placing top 5 nationally.

**DUM E Robotic Arm Project | Sep 2026 - present**

- Contributing to the mechanical build of an SO-101-based robotic arm (~3 hrs/week): CAD, assembly, and testing.
- Own funding/sponsorship outreach and team admin/GitHub for a 3-person build team.

**Quant Strategy Prototypes - Itau Desafio Quant AI 2026**

- Built three self-contained quant strategy prototypes in Jupyter: prediction-market signals on B3 assets, an XGBoost regime classifier on IBOV returns, and an event study of sports results on sponsor stock returns, with Sharpe and max-drawdown backtests.

## PROJECTS

**Eye-Tracker - Personal Project**

- Building a webcam-based gaze-tracking control layer: full hands-free computer control with zero keyboard or mouse input.
- Designed the gaze-plus-voice interaction model: open-palm gesture as push-to-talk, voice-only destructive confirmations, gaze/voice guided first-run tutorial.

## SKILLS

Python, C++, PyTorch, TypeScript, MATLAB, Fusion 360, SolidWorks, Git, Linux, EasyEDA, Arduino, ESP32, Raspberry Pi

## AWARDS

Bronze, Brazilian Mathematics Olympiad | Silver, Mathematique Sans Frontieres | People's Choice, U of T Undergraduate Engineering Research Day (UnERD 2026)